

# Computer Graphics.

## Unit - I

⇒ Computer Graphics:- Computer graphics may be defined in one of the following ways:-

→ Definition - I:- CG is the discipline of producing pictures or images using a computer including

- \* Modelling →
  - creation
  - manipulation
  - Storage of geometric objects

and \* Rendering →

- converting a scene to an image
- process of transformation
- Animation of the image.

→ Definition - II:-

Computer graphics = Graphics algorithm + Data Structure + Language.

- data structure suitable for Computer graphics.
- Algorithms suitable for picture generation and transformations.
- An appropriate language construct for the generation of graphic objects.

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⇒ Computer Graphics Applications :- various applications of computer graphics are as follows:-

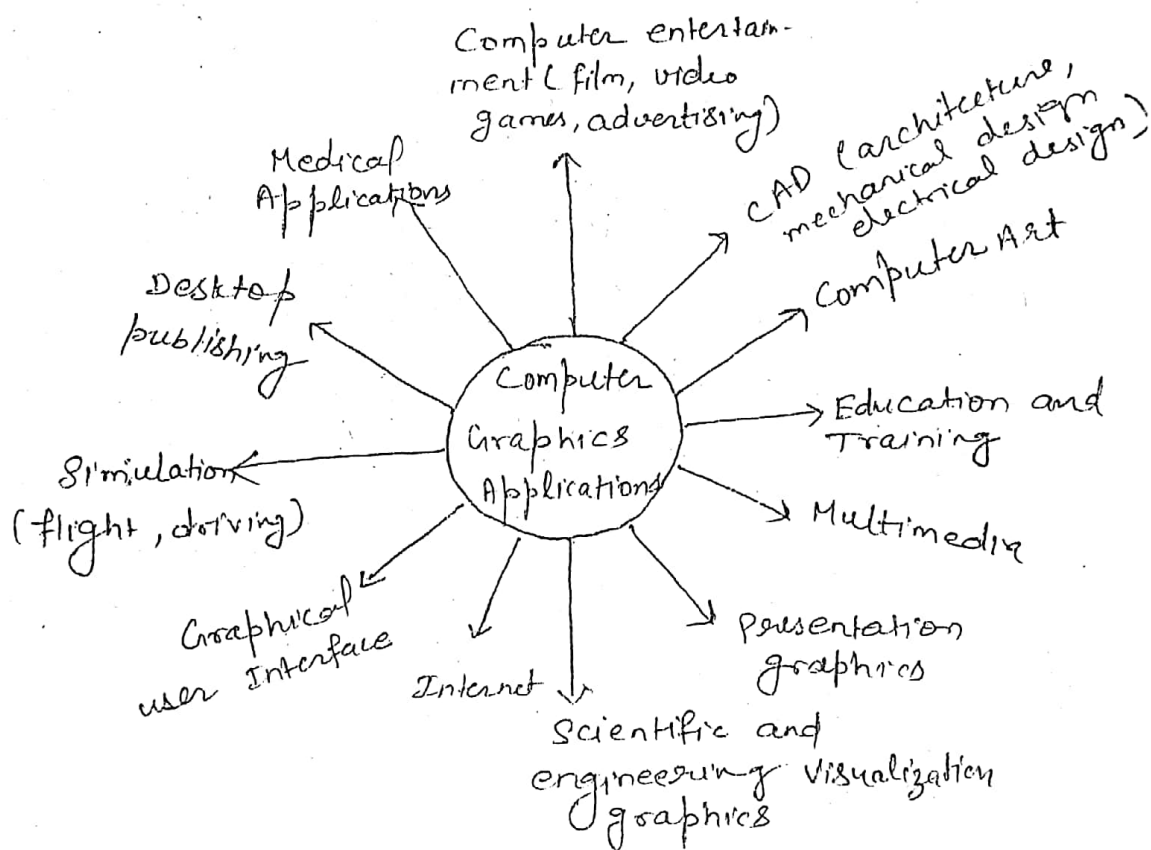


Fig:- Applications of Computer Graphics.

- 1) Computer Aided design (CAD) :- This is particularly used in engineering applications such as
- building and other structural design
  - Industrial design
  - Design of manufacturing process
  - automobiles and aircrafts.
  - Ships and spacecrafts.
  - very large scale integrated chips (VLSI)
  - And telephone and computer network

(3)

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The design process involves modelling of an object or conceptual model of a process on which modification, analysis and other manipulations can be carried out.

- 2) Computer Art :- In earlier time a painter spend energy and the material (paints, pencil, paper etc.) for making a painting, but after the use of computer graphics, he don't need a paper and pencil for making a drawing.
- He can erase and modify his drawing any number of times for the better results.
  - cartoon pictures, stylish advertising text, logos are few of the example of the use of graphics in commercial arts.

### 3) Education and Training :-

- In modern education systems, there is a huge application of CG in the form of models of physical, economical and mathematical systems.
- Trainees in different areas like medical, engineering etc. takes help of graphical models for easy understanding of the subject.
- For a trainer also it is easy to communicate his ideas to the follower in the form of

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graphics and he does not need to use the verbal or complicated text for communication.

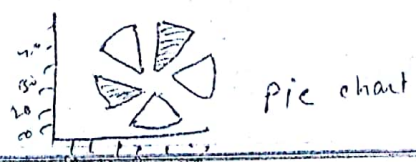
- An education using graphical method can be very useful for a multilingual (using different languages) groups where verbal or text may not be used to deliver exact ideas.

4) Multimedia :- Multimedia is the field concerned with the computer-controlled integration of text, graphics, drawings, still and moving images (video), animation, audio can be represented, stored, transmitted and processed digitally.

5) Presentation Graphics:-

- Presentation graphics are basically used to summarize the scientific, mathematical, business statistical data for report purpose.
- These reports can be management report, research reports etc.
- Example of presentation graphics are:
  - line graph
  - pie chart
  - bar chart etc.

which shows the relationship b/w different parameters.



## 6) Scientific & Engineering Visualization:-

- In today's world of science, visualisation is not just a means of presenting the data but is very essential in order to extract meaning and information.
- visualisation is a means of scientific enquiry and research which is an integral part the scientific process.
- Computer graphics helps the visualization application.

7) Internet :- The word Internet is derived from two words -

- Interconnection and
- Networks

→ The internet would be positively boring without graphics

## 8) Graphical user Interface (GUI):-

- In traditional approach the user performed their tasks by typing command but now the GUI make the task easy and more interactive by putting all the command in the form of menus and icons.
- Major components of GUI are menus, icons, and multiple window areas.
- The main advantage of these systems are that they take less screen space and easy to use and learn.

(6) → the act of giving a false appearance (बनावट)  
9) Simulation and Virtual Reality :- with the help of simulation, the operator would see the views under various circumstances.

- Complex, mechanical, chemical and industrial processes can be simulated in action so that a person can be trained in his operation before they handle the actual equipment. Such a practice avoids wastage of cost and time.
- An airplane pilot trained in an imaginary airplane for number of times before going to have the actual flight.
- Simulation is also useful in other difficult applications like nuclear testing.

10) Desktop Publishing (DTP) :- DTP is related to the printing newsletters, book publishing in small organization.

11) Medical Applications :-

- Computer graphics is used in biology to demonstrate the internal parts of body through X-ray or ultra sound.
- Other uses of computer in biology are molecular study, ECG, MRI etc.
- CG is used to simulate the complex biological activities like heart pumping, digestive.

System working etc.

12) Computer games and entertainment :- Computer graphics can be used broadly in the field of entertainment.

- CG are used in making motion pictures, music video etc.
- In our latest movies there are lots of graphics used to give special effect in the movies (animated movies).
- Modern video games are also based on graphics where lots of imaginary characters and situations developed through CG.

⇒ Computer graphics H/w and S/w (or overview of graphic system or devices) :-

→ Computer graphics Hardware :-

\* Input devices :- various devices are available for data input on graphics workstations. These are described below :-

1) Keyboards :- An alphanumeric keyboard on a graphics system is used primarily as a device for entering text strings.

- keyboards can also provided with features to facilitate entry of screen coordinates, menu selections or graphics functions.

2) Mouse :- A mouse is small hand held box used to position the screen cursor.

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- wheels or rollers on the bottom of the mouse can be used to record the amount and direction of movement.

3) Trackball :- As the name implies, a ball that can be rotated with fingers or palm of hand to produce screen cursor movement.

- Potentiometer attached to the ball measures the amount and direction of rotation.

4) Joystick :- A joystick consists of a small, vertical stick mounted on a base that is used to move the screen cursor around.

- One or more buttons can be programmed to act as I/O switches to signal certain action once a screen position has been selected.

5) Image Scanner :- Drawing graphs, coloured and black & white photos or text can be stored for computer processing with an image scanner by passing an optical scanning mechanism over the information to be stored.

6) Touch panel :- It allows displayed objects or screen positions to be selected with the touch of a finger.

7) Voice System :- Speech recognizers are used in some graphics workstations as input devices to accept voice commands.

⑦) \* Output devices :- Primary output devices in the graphics system is a video monitor.

The operation of the most video monitors is based on the standard cathode ray tube (CRT) design.

1) CRT :- CRT is an evacuated glass tube containing an electron gun and fluorescent screen.

→ Components of CRT :-

- Electron gun :- It consists of a heated metal element and a cathode.
- Control grid :- It is used to control the intensity of the electron beam generated from gun by setting voltage levels.
- Focusing System :- It is used to focus the electron beams as a small spot on the phosphorus screen.
- Deflection System :- It is used to control the direction of electron beam by generating magnetic or electric field.
- Phosphor Coated screen :- The inner surface and (front) is coated with phosphor. Phosphor converts the electrical energy into light energy. when electron beam strikes on the phosphor screen, screen emits light.

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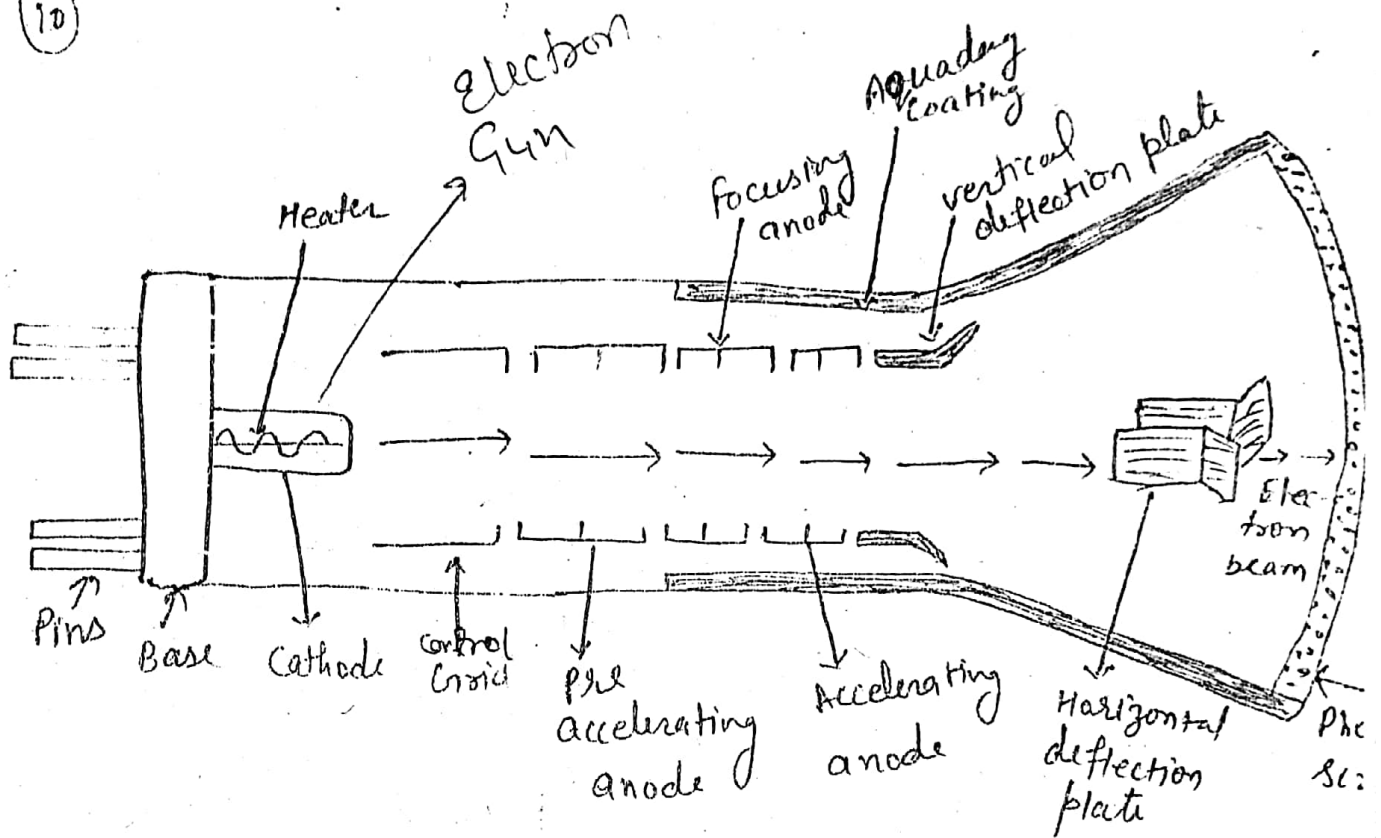


Fig:- Cathode Ray Tube

Operation of CRT:- Working of CRT is simple and based on the concept of electron beam propagation.

\* When current is supplied to the heating element of electron gun, it gets heated and the cathode also gets heated, this causes the electrons to emerge out and pass through the cathode ray tube towards the phosphor coated screen. These electrons are negatively charged.

\* Now by using the control grid, the intensity of the electron beam can be set at a desired level by setting the voltage level.

\* Pre accelerating anode slowly accelerates the electrons up to some level.

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(11) Now when electron beam passes through grid and preaccelerating anode, it spreads out, now focusing system is used to convert the electron beam as a narrow beam to converge into a small spot on the screen.

Focusing can be done by electric or magnetic field developed by magnetic focusing coil.

- \* Accelerating anode accelerate the narrow beam of electrons.

- \* Direction of electron beam can be controlled by deflection system.

Electric or magnetic fields are used in deflection system.

- \* When electron beam falls on the phosphor coated screen that part of the screen begins to emit light and picture can be displayed on the screen.

- \* Aquadag coating is used to maintain the electrical equilibrium (i.e. when there is huge amount of electron striking to the phosphor screen and also collide electrons with other electrons then aquadag coating absorbs them and maintain the electrical equilibrium)

- \* Since when the electron beam stopped, picture will destroy because the light emitted by.

(12) phosphor fades out very soon so some technique is needed to retain the picture on the screen.

In this picture definition is stored in a memory area called refresh buffer or frame buffer and this type of CRT is called Refresh CRT.

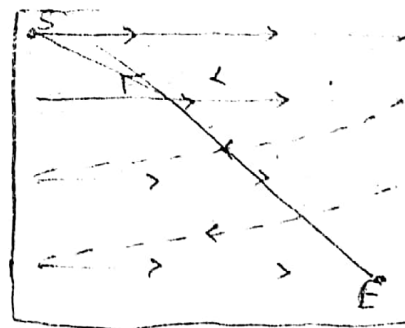
→ There are two technologies used in graphics system —

\* Raster scan display

\* Random Scan display [or Vector Scan]

Raster Scan System :- In a raster scan system, the  $e^-$  beam is swept across the screen, one row at a time from top to bottom.

- As the  $e^-$  beam moves across each row, the beam intensity is turned on and off to create a pattern of spots.
- Picture definition is stored in a memory area called refresh buffer or frame buffer.
- The raster scan proceeds as follows - Starting from the top left corner of the screen, the electron gun scans horizontally from left to right, one scan line, that is, one row at a time, jumping (without tracing the line) to the left end of the



Raster Scan cycle  
(S = start, E = End)

- (13) next lower row until the bottom right corner is reached. Then it jumps again without tracing to the top left corner and starts again, finishing one complete refresh cycle.
- Each screen point is referred to as a pixel or pel.
- Example of HWT - <sup>referred</sup>

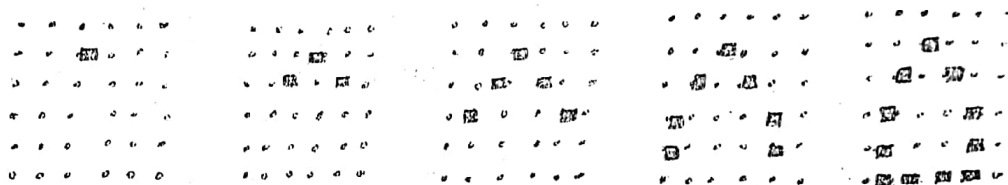


Fig- Raster Scan display of hwt.

#### Advantages:-

- 1) Picture quality is <sup>not so</sup> good.
  - 2) very cost effective.
  - 3) Large memory space.
  - 4) High Refreshing Speed. (60-80 frame/second)
  - 5) zig-zag line will produce
- 6) use pixels to draw an image
- 7) use frame buffer for storage

Random Scan Display:- when operated as a random scan display unit, a CRT has the e<sup>-</sup> beam directed only to the parts of the screen where a picture is to be drawn.

Random Scan

- Random scan monitors draw a picture one line at a time and for this reason it is also referred to as vector display.
- In this efficiency is achieved by minimizing wasted motion as would happen if a line segment starts at a point different from the

(14) end point of the previous segment.

Example of Hut -

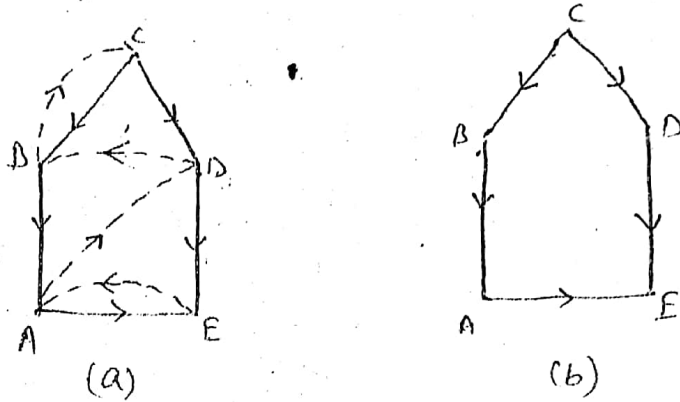


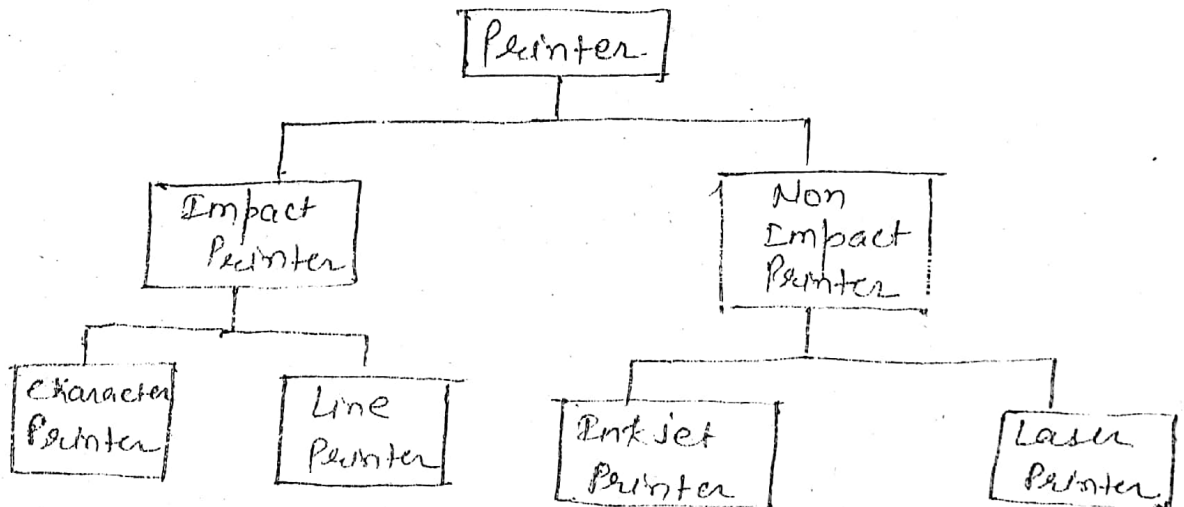
Fig - vector scan display

(a) CB - CD - DA - DE - AE

(b) ABCDEA

(- - - - Beam off, — Beam on)

2) Printers - Printer produce o/p by either impact or non impact method.



Impact Printer:- Impact printer prints the character by striking against inked ribbon against on the paper.